Classical Simulation of Two-Layer Quantum Circuits via Output-Dependent Cubic Structure

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Abstract

We give a constructive method for estimating one output probability of a specified two-layer Boolean cubic-phase circuit. Differentiating the second phase at the measured output produces a quadratic Walsh support. Restricting the first cubic tensor to its displacement space yields a control parameter that can remain constant even when specified global-control certificates grow. We construct normalized matched sectors from the source and apply established stabilizer norm estimation to their coherent sum. For a verified restricted control codimension k, the readout uses $2^k\epsilon^{-2}\log(1/\delta)$ times a fixed polynomial, compared with 4^k for direct coherent branch pairs. General control discovery still requires a full-rank contraction-anchor promise and costs $2^{O(q^2)}$ at cap q; deterministic shortcuts have separate guarantees. We give exact arithmetic, random-bit and storage bounds, growing examples, explicit refusals, and a reproducible finite comparison that preserves the new method's unfavorable small-k constants. This is a source-specific application and refinement of existing covering-set, independent-space and norm-estimation methods, not a new general simulation primitive or a universal quantum simulation result.

1 Task, source interface and contribution

Let $V = \mathbb{F}_2^n$, let J be a symmetric invertible binary matrix, and define

$$(F_J)_{x,y} = 2^{-n/2}(-1)^{x^T Jy}, \qquad D_f|x\rangle = (-1)^{f(x)}|x\rangle.$$

Thus $F_J = F_J^* = F_J^{-1}$. The source describes

$$|\Psi\rangle = F_J D_{f_1} F_J D_{f_0} F_J|\beta\rangle, \qquad p = \Pr[e^T x = 1 \text{ on measuring } \Psi], \tag{1}$$

where f_0, f_1 have Boolean degree at most three, β is a specified basis input, and e is a specified coordinate vector. All polynomials are reduced algebraic normal forms: $x_i^2 = x_i$, coefficients lie in $\mathbb{F}_2$, and constant, linear and quadratic terms are retained. We estimate p within $\epsilon = 2^{-b}$ with failure at most 2^{-h}, for integers $b, h \geq 1$. This is one marginal probability, not full-distribution sampling.

The executable native interface is deliberately literal. A descriptor contains n, J, f_0, f_1, β, e as explicit binary masks. The native gate list begins with an identity header encoding the descriptor: each ASCII bit is a doubled X or doubled Z on wire zero, terminated by a doubled H. The body is the canonical expansion of (1): reversible linear maps and Hadamards implement F_J; Z, controlled-Z and the fixed Clifford+T decomposition of controlled-controlled-Z implement the

phase monomials. Constant signs use $XZXZ$. Recognition decodes the header, validates the descriptor, recompiles it, and compares every body record and boundary value. Hence the header supplies an encoding of the actual source, not an unchecked claim of equivalence or a structural certificate. An arbitrary equivalent gate list need not be recognized. The theorem also applies directly to the explicit descriptor interface with its actual encoding cost. Let B be the bit length of the chosen literal source, including its header when present.

The source contains neither a control space nor a stabilizer decomposition. Those are constructed and verified. The contribution is the resulting original-source theorem: output-dependent sector construction, a promised discovery procedure, and a joint norm readout with all access and arithmetic costs included. The conditional quadratic decompositions are instances of established covering and independent-space methods [3, 1]; the linear-length norm estimator is established [2]. Output-dependent marginal access is also prior art [4, 5]. Section 7 identifies the interfaces precisely.

2 Quadratic output reduction and normalized sectors

Put $j = J^{-1}e$ and $Q(v) = f_1(v)+f_1(v+j)$. The degree of Q is at most two. For a quadratic q, write $B_q(u,v) = q(u+v)+q(u)+q(v)+q(0)$ for its alternating polar form and $W_q(l) = \sum_v (-1)^{q(v)+l^T v}$ for its Walsh transform.

Lemma 2.1 (Quadratic Walsh elimination)**.** *If B_q has rank r, then r is even. The support of W_q is an affine space $l_0 + \operatorname{im} B_q$ of dimension r. Every nonzero normalized value $2^{-n}W_q(l)$ has magnitude $2^{-r/2}$, and its sign is a quadratic function of coordinates on this support. The support, its origin and this sign can be computed by binary elimination.*

Proof. Choose a basis in which the polar form is a sum of $r/2$ hyperbolic pairs and has a radical of dimension $n-r$. In those coordinates q is a sum of pair products plus linear terms and a constant. Summing a radical coordinate is zero unless its linear coefficient is canceled by l, and otherwise contributes two. Each pair sum is $\sum_{u,v}(-1)^{uv+au+bv} = 2(-1)^{ab}$. Multiplying gives the stated support, magnitude and quadratic sign, with no enumeration of support points. The same elimination computes an exact real quadratic Gauss sum in polynomial bit time. □

Apply Lemma 2.1 to Q. Write its support as $J(d_0+S)$, where $S \subset V$ has even dimension r. Choose a basis matrix $D : \mathbb{F}_2^r \to S$ and a computed quadratic w with

$$2^{-n}W_Q(J(d_0+Dz)) = 2^{-r/2}(-1)^{w(z)}. \tag{2}$$

Let $C : V \to \mathbb{F}_2^c$, $c = n-r$, be a full-row-rank annihilator of S, let R be a right inverse of C, put $\eta = Cd_0$, and solve $Ds = d_0 + R\eta$. These are all computed affine coordinates, including the origin and its matching offset.

The first phase state has amplitudes $2^{-n/2}(-1)^{f_0(x)+(J\beta)^T x}$. Its mass on each coset $Cx = a$ is exactly 2^{-c}. In coordinates on that coset define

$$\psi_a(y) = 2^{-r/2}(-1)^{g_a(y)}, \qquad g_a(y) = f_0(Ra+Dy) + (J\beta)^T(Ra+Dy). \tag{3}$$

Every ψ_a is a unit vector.

Proposition 2.2 (Matched-sector identity)**.** *The pulled-back output observable is the real Hermitian Clifford unitary*

$$O = F_J D_{f_1} F_J Z_e F_J D_{f_1} F_J = F_J D_Q F_J Z_e.$$

It maps coset a onto coset $a+\eta$. Its reduced kernel is

$$O_a(t,y) = 2^{-r/2}(-1)^{q_a(y,t)}, \quad q_a(y,t) = w(y+t+s) + e^T(Ra+Dy). \tag{4}$$

The cross bilinear matrix of q_a in (y,t) is nonsingular. Moreover,

$$X_a = \langle \psi_{a+\eta}, O_a\psi_a\rangle \in [-1,1], \quad X_a = X_{a+\eta}, \quad \mathbb{E}_a X_a = 1-2p. \tag{5}$$

For $u_a = \psi_{a+\eta} + O_a\psi_a$, we have $\|u_a\|^2 = 2+2X_a \in [0,4]$.

Proof. $F_J Z_e F_J$ is translation by j. Commuting this translation through D_{f_1} gives D_Q, proving the second expression for O. Quadratic phase gates and binary Fourier maps are Clifford. The first expression proves Hermiticity and unitarity. The matrix entry between x and x' is $2^{-n}W_Q(J(x+x'))(-1)^{e^Tx}$. Its support condition is $x+x' \in d_0+S$. Substitute $x = Ra+Dy$, $x' = R(a+\eta)+Dt$ and use (2) to obtain (4). Since $Q(v+j) = Q(v)$, its Walsh support annihilates j; thus e annihilates S and $e^Td_0 = 0$. Each restricted map is a square unitary between full cosets. In a flat quadratic kernel, a singular cross matrix would give two proportional rows, contradicting unitarity. Realness and Cauchy–Schwarz bound X_a; Hermiticity gives its matching symmetry. Expanding the first phase state over orthogonal cosets of equal mass gives (5). Expanding the norm of u_a proves the last assertion. All lower terms and input characters remain in these formulas. □

Equivalently, substitute $x = Ra+Dy$ and displacement d_0+Dz in the full output polynomial. Its signed numerator H_a has $2r$ free variables and $X_a = H_a/2^{3r/2}$. This smaller normalization matters: summing terms independently before retaining their interference would not provide the bounded statistic in (5). There is no unknown postselection probability to estimate.

3 Control certificates and paid discovery

The third difference of a Boolean cubic is an alternating trilinear form. Let T be this tensor for f_0 restricted to S in the computed D coordinates. A control certificate is a k-dimensional covector space $U \subset (\mathbb{F}_2^r)^*$ with $K = \ker U$ and $T(K,K,K) = 0$. This is an independent-space condition in the terminology of [1]. It can be checked by tensor evaluation on a basis of K. Translation and addition of lower-degree terms do not change T.

Lemma 3.1 (Conditional quadratic branches)**.** *A verified control certificate constructs a decomposition of each ψ_a into 2^k normalized stabilizer states, each with coefficient $2^{-k/2}$. Fixing controls on both sides of X_a yields 4^k exact real quadratic sums.*

Proof. Choose a right inverse of U and a basis of K, yielding coordinates (u,z) on $\mathbb{F}_2^r$. On each affine slice u the cubic coefficient is $T|_{K^3} = 0$, so the full phase, including its affine terms, is quadratic in z. A uniform quadratic phase on an affine subspace is a normalized stabilizer state. The slice has 2^{r-k} points, which gives coefficient $2^{-k/2}$ in (3). The kernel in (4) is quadratic, so fixing both state controls leaves a quadratic form on $2r-2k$ variables. This is the established covering-set construction in a computed coordinate frame, not a minimum stabilizer-rank claim. □

We use the following explicit discovery scope, with integer search cap $q \geq 0$. Zero tensor is accepted deterministically. A nonzero one-control factor is detected by solving $\ell \wedge T = 0$ for a covector ℓ: in a basis with $\ell = e_1^*$, this equation says every cubic includes that coordinate. Thus a nonempty solution space supplies a control, and an empty space is a valid cap-one refusal. At larger caps a basis of the quadratic exterior kernel is scanned for decomposable two-forms; every proposed image is verified. This is sound but incomplete: a decomposable linear combination need not be one of the returned basis vectors.

Theorem 3.2 (Capped anchor discovery)**.** *Suppose some certificate has dimension $k \le q$ and some $v \in K$ has contraction $A(v) = T(v,\cdot,\cdot)$ of rank $2k$. After deterministic shortcuts and coordinate probes, inspect at most*

$$A_q = 4^q(h+2) \quad \textit{uniform probes}, \qquad C_q = 4(q+1)2^{q^2} \quad \textit{candidates per image}. \tag{6}$$

Stream all covector subspaces of dimension at most q in each nonzero contraction image of rank at most $2q$, verifying each. Discovery misses with probability at most 2^{-h-1}. Every accepted certificate is sound, even without the promise. Every random tape obeys the caps; a miss outside the promise has no success guarantee.

Proof. For $v \in K$, the contraction vanishes on $K \times K$. In coordinates complementary to K its matrix has block form $\left(\begin{smallmatrix} B & M \\ M^T & 0 \end{smallmatrix}\right)$. Rank $2k$ forces M to have row rank k, and consequently the contraction image contains U. A principal Pfaffian witnessing rank $2k$ is a degree-k polynomial in $v \in K$. It is nonzero at the promised anchor. Any nonzero Boolean polynomial of degree at most k is nonzero on at least a 2^{-k} fraction of points: induct on the degree, using a nonzero highest-variable derivative and one nonzero value in each corresponding pair. A uniform ambient vector hits K with probability 2^{-k}. Thus each probe is useful with probability at least $4^{-k} \ge 4^{-q}$.

The number of d-subspaces in an l-space is a Gaussian binomial coefficient, at most $4\,2^{d(l-d)}$; indeed its product formula has denominator $\prod_{i=1}^{d}(1-2^{-i}) > 1/4$. For $l \le 2q$ and $0 \le d \le q$, $d(l-d) \le q^2$, proving the stated candidate cap. An image from a useful probe therefore exposes U to the exhaustive verified search. Independent probe failure is at most $(1-4^{-q})^{A_q} \le e^{-(h+2)} \le 2^{-h-1}$. Coordinate probes and sound shortcuts can only improve success. Rank above $2q$ is skipped, not treated as impossibility. The final complete phase substitution checks the expected absence of free cubic terms as an additional consistency check. □

A user-imposed candidate limit that truncates this search removes its confidence guarantee. Small-control existence alone is insufficient for the anchor theorem. The algorithm returns explicit discovery or grammar refusals; it has no hidden unbounded fallback and makes no generic gate-equivalence recognition claim.

4 Joint sector and equatorial norm sampling

By Lemma 3.1 and the Clifford property, u_a has at most 2^{k+1} constructed stabilizer terms. The established norm estimator of Bravyi et al. [2, Section 4.3] is therefore applicable with work linear in that length. We include its elementary moment argument to make our normalization and joint sampling contract explicit.

Choose independent $d_j \in \mathbb{Z}_4$ uniformly and independent fair bits a_{jl}, and put

$$z_A(t) = \sum_j d_j t_j + 2\sum_{j<l} a_{jl} t_j t_l \pmod 4, \qquad \phi_A(t) = 2^{-r/2} i^{z_A(t)}. \tag{7}$$

This requires exactly $d_r = 2r + r(r-1)/2$ fair bits.

Lemma 4.1 (Equatorial moments; established norm primitive)**.** *For any complex vector u indexed by $\mathbb{F}_2^r$, set $Z = 2^r|\langle \phi_A, u\rangle|^2$. Then*

$$\mathbb{E}Z = \|u\|^2, \qquad \mathbb{E}Z^2 = 2\|u\|^4 - \sum_t |u(t)|^4, \qquad \operatorname{Var} Z \le \|u\|^4.$$

Proof. The independent diagonal phases kill every off-diagonal term of the first moment. For the fourth-order expansion, a surviving tuple $(x, z; y, w)$ must obey $x_j + z_j = y_j + w_j$ as integers at every coordinate: the difference lies in $[-2, 2]$ and must vanish modulo four. Where x, z differ, (y_j, w_j) is either their original order or its swap. If two differing coordinates have different swap choices, the associated independent edge phase contributes an odd sign and kills the term. All choices must therefore agree, leaving exactly the two pairings $(y, w) = (x, z)$ and (z, x). Their overlap $x = z = y = w$ was counted twice and must be subtracted. This gives the second-moment identity, including $r = 0$ and $u = 0$. □

For $u = u_a$, exact access uses

$$A = \sum_t i^{-z_A(t)}(-1)^{g_a+\eta(t)},$$
$$B' = \sum_{y,t} i^{-z_A(t)}(-1)^{g_a(y)+q_a(y,t)},$$
$$Z_{a,A} = \frac{|2^{r/2}A + B'|^2}{2^{2r}}. \tag{8}$$

Fixing k controls evaluates A with 2^k quadratic $\mathbb{Z}_4$ sums on $r - k$ variables and B' with 2^k such sums on $2r - k$ variables. There is no 4^k branch fallback in this readout. For a quadratic $\mathbb{Z}_4$ form with even cross coefficients, let I index its odd linear coefficients and $L(x) = \sum_{j\in I} x_j \pmod 2$. The identity

$$\sum_{j\in I} x_j \equiv L(x) + 2\sum_{j<l\in I} x_j x_l \pmod 4$$

reduces the sum, up to a constant power of i, to $(G_0 + G_1)/2 + i(G_0 - G_1)/2$, where G_0, G_1 are real quadratic sums differing by L. Affine substitution preserves this form. Thus each joint draw uses exactly $4 \cdot 2^k$ real Gaussian calls. Because r is even, (8) has an integer numerator and dyadic denominator. No floating-point cancellation or separate accuracy budget for a norm oracle is needed.

Theorem 4.2 (One accuracy budget)**.** *Sample a uniformly from matching-orbit representatives and A independently as in (7), and set $Y = Z_{a,A}/2 - 1$. Then $\mathbb{E}Y = 1 - 2p$ and $\operatorname{Var} Y \le 4$. With*

$$L = 32\,4^b, \qquad K_g = 2h + 3, \qquad N = LK_g, \tag{9}$$

take the median of K_g independent group means of L samples each. Convert it to a probability, project the final value to $[0, 1]$, and round down at $b + 2$ fractional bits. Conditional on sound discovery the returned value differs from p by at most $3\epsilon/4$, except with probability less than 2^{-h-1}. On the promise of Theorem 3.2, total discovery/readout failure is less than 2^{-h}.

Proof. The orbit symmetry in (5) permits representatives sampled with c bits if $\eta = 0$, or $c - 1$ bits otherwise (fix a coordinate where η is one). Lemma 4.1 gives $\mathbb{E}[Y \mid a] = X_a$ and $\operatorname{Var}(Y \mid a) \le (1 + X_a)^2$. For $\mu = \mathbb{E}X_a$, total variance is bounded by

$$\mathbb{E}(1 + X)^2 + \operatorname{Var} X = 1 + 2\mu + 2\mathbb{E}X^2 - \mu^2 \le 4 - (1 - \mu)^2 \le 4.$$

Chebyshev bounds a group's error probability at tolerance ϵ by $1/8$. For the median to fail, at least half the independent groups must fail, so a union bound over subsets gives $2^{K_g}(1/8)^{K_g/2} = 2^{-K_g/2} < 2^{-h-1}$. Probability conversion divides the error by two; projection cannot increase it; rounding adds at most $\epsilon/4$. A union bound adds discovery failure. The conditional argument is uniform over accepted frames, even those selected randomly by discovery. Zero norms cause no exception: then $Y = -1$ identically. □

Individual Y values are not bounded by one. Applying bounded-variable Hoeffding to them or clipping individual samples would be invalid. Only the final probability is projected. The retained coherent evaluator instead samples the genuinely bounded X_a with $2(h+2)4^b$ draws and uses 4^k Gaussian calls per draw.

5 Complete resource contract

Let $m = n + r$, and use A_q, C_q, N, K_g from (6) and (9). With elementary dense substitution and binary elimination, a conservative bound on every random tape, including refusals, is

$$\begin{aligned}\text{bit time} = \widetilde{O}\big((B+m+q+2)^{12} + (r+A_q)(1+C_q)(m+q+2)^{10} \\ + N2^{k+2}(m+q+2)^{10} + (N+K_g+1)(m+b+h+q+2)^4\big).\end{aligned} \tag{10}$$

Here k is the returned dimension; $k \le q$ supplies an a priori bound before discovery. The source term pays decoding, native replay, tensor reconstruction, Walsh support, exterior kernels and all coordinate construction. A cubic has $O(m^3)$ coefficients and its explicit linear substitutions can be performed within $O(m^6)$ elementary binary operations; solving the exterior systems and verifying candidate kernels fits the displayed loose polynomial powers. Each probe uses binary matrix elimination; each candidate adds tensor and complete phase checks. The third term pays all affine restrictions, random graph construction and the real Gaussian calls for the two widths in (8). These powers are upper bounds for a straightforward implementation, not fitted performance exponents.

Integers in a quadratic sum and a norm numerator have $O(m)$ bits. Group accumulation adds $O(\log L)$ bits, and schedules have $O(b + \log(h+2) + q)$ bits. Keeping group totals while streaming probes, candidates and branches gives the conservative bounds

$$\text{space} = \widetilde{O}\big((B+m+q+2)^8 + (m+b+h+q+2)^4\big), \tag{11}$$

$$\text{fair bits} \le rA_q + N(c - \mathbf{1}_{\eta\neq 0} + 2r + r(r-1)/2). \tag{12}$$

Deterministic shortcuts omit the anchor term. Candidate storage is polynomial because subspaces are enumerated and checked one at a time. Samples and Gaussian budgets are checked before any readout draw. Truncation yields a refusal, not a success claim.

If the complete original displacement phase is already quadratic, a retained exact shortcut uses one real Gaussian sum, no sampling bits and polynomial cost in b. Otherwise sampled evaluation is polynomial in $1/\epsilon$, and exponential in b. On the anchor promise $q^2 = O(\log(B+m+2))$ suffices for polynomial total inverse-error cost. Improving 4^k to 2^k in readout does not improve the $2^{O(q^2)}$ discovery term. Neither low restricted rank nor a favorable anchor is asserted for every cubic.

6 Growing families and scope limits

For even $t \ge 2$, take coordinates $c_0, w_0, X \in \mathbb{F}_2^{t\times t}, U, V \in \mathbb{F}_2^t$, so $n = t^2 + 2t + 2$, and put

$$\begin{aligned} f_0 &= c_0 w_0 + \sum_{i,j} X_{ij}U_iV_j, \\ f_1 &= c_0 w_0 + c_0(Q_{\det}(X) + U_0V_0), \\ Q_{\det}(X) &= \sum_{i,j \text{ even}} (X_{ij}X_{i+1,j+1} + X_{i,j+1}X_{i+1,j}). \end{aligned} \tag{13}$$

Initially $J = I$, $\beta = 0$, and the output is c_0.

Proposition 6.1 (Restricted versus specified global controls)**.** *For* (13)*, S is the span of X, U_0, V_0, $r = t^2 + 2$, and $k = 1$ is detected by the deterministic one-control test. Every global certificate $T(K_0, K_0, K_0) = 0$ of codimension g for the full first cubic satisfies $g \geq t - \sqrt{t}$. The output is*

$$p = \frac{1}{4} - 2^{-t-1} + 2^{-2t-1} < \frac{1}{4}. \tag{14}$$

Proof. The derivative of f_1 is $w_0 + Q_{\det}(X) + U_0 V_0$. Its polar is nonsingular on the stated S and zero on its complement, proving the support direction and rank. Restriction of the first cubic to S leaves $X_{00} U_0 V_0$, which has one control. For the global bound assume $g < t$ (otherwise it is immediate). Intersections of K_0 with the U and V coordinate spaces have dimensions at least $t - g$; its intersection with the matrix space has codimension at most g. Tensor vanishing forces $u^T X v = 0$ on those intersections. Their tensor product has dimension at least $(t-g)^2$ and lies in the annihilator of $K_0 \cap \mathbb{F}_2^{t \times t}$, of dimension at most g. Hence $(t-g)^2 \leq g \leq t$ and $g \geq t - \sqrt{t}$.

For the probability, Fourier summation over X in the first phase forces the frequency matrix to equal UV^T for some U, V. Thus every nonzero spectral amplitude has a rank-at-most-one frequency matrix, on which $Q_{\det}$ vanishes. The c_0, w_0 Fourier pair cancels the derivative's w_0 sign in the shifted spectral overlap. The remaining expectation is the squared Fourier-amplitude average of $(-1)^{l_{U,0} l_{V,0}}$. Expanding $(-1)^{ab} = (1 + (-1)^a + (-1)^b - (-1)^{a+b})/2$ gives four characters. By Fourier orthogonality each character average equals the first phase state's autocorrelation under the corresponding physical translation. The zero translation contributes one. Translation of U_0 alone requires $V = 0$ and contributes 2^{-t}; translation of V_0 alone contributes 2^{-t}. Translation of both requires $U_i = V_i = 0$ for $i > 0$ and $U_0 + V_0 = 1$, contributing 2^{1-2t}. On every surviving matrix the determinant blocks vanish. Thus $1 - 2p = (1 + 2^{1-t} - 2^{1-2t})/2$, which is (14). □

Adding the output linear term to f_1 complements p above $3/4$. Arbitrary lower terms in f_0 preserve both tensor certificates. Lower terms in f_1 change its derivative only by an affine term, preserving S but possibly changing d_0, η. Basis inputs add linear characters. Invertible physical coordinate changes preserve the certificate conditions with the Fourier pairing transformed consistently. These variations remain covered even when (14) no longer applies. One can also add first-phase cubics whose tensor vanishes on S^3, preserving small restricted k; the global lower bound is not asserted for every such extension.

The global bound concerns this control certificate, not unrestricted stabilizer rank or all classical algorithms. In fact the unperturbed formula (14) is cheaper than either sampler. Restricted controls can also grow: for independent variables Y_{ij} and d_i, the cubic $\sum_{i<j} Y_{ij} d_i d_j$ requires $\binom{t-k}{2} \leq k$ for any codimension-k independent space. To see this, project that space onto d (dimension at least $t-k$) and intersect it with Y (codimension at most k). Tensor vanishing places the exterior square of the projection in an annihilator of dimension at most k. Thus $k \geq t - \sqrt{2t}$ is necessary. This delimits the certificate mechanism; it is not a hardness theorem.

7 Relation to prior simulation interfaces

Maslov et al. [3, Section II] condition on covers of cubic variables to obtain quadratic sums. Lemma 3.1 is that construction after a computed affine change of coordinates. Bläser et al. [1, Definition 1 and Theorem 41] use the same independent-space condition and evaluate an amplitude with a supplied $(r-k)$-space in $O(2^k r^3)$ work. Their Theorem 67 has recovery work $O((r-k) r^3 2^k)$ and high probability over a random planted model when the planted dimension exceeds half the

Specified method	Readout factor	Access obligation
Normalized Pauli expansion	2^r	Quadratic Walsh support
Global independent space plus norm sampling	2^g	Find or supply global controls
Coherent matched-sector pairs	4^k	Restricted discovery
Joint sector/norm sampling	2^k	Same discovery, kernel construction
Unperturbed family formula	polynomial	Recognize that special family

Table 1: First four factors multiply $\epsilon^{-2}\log(1/\delta)$ and their actual source/arithmetic polynomials. Discovery is additional. The last method is exact. These bounds neither share a machine-level cost unit nor imply a wall-time ordering.

ambient dimension by a fixed fraction. Our slower anchor search gives a per-promised-input guarantee with a tunable failure budget. These are different quantifiers; we do not improve their planted recovery bound or claim discovery from small-control existence alone.

Park et al. [4, Supplement, Section IV.3, Theorem 4] give framed-Wigner marginal methods for Clifford circuits with specified product inputs and positive representation access. The first cubic phase state here need not be a product: the CCZ phase has a flattening minor $\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$ across one qubit versus the other two. Moving that phase into the operation introduces a non-Clifford gate. This identifies a missing hypothesis for the literal reduction; it does not exclude other embeddings with explicitly paid access costs.

The first phase state is computationally tractable in the sense of Van den Nest [5, Sections 4.1 and 4.3]: its amplitudes are evaluable and its squared magnitudes are uniform. However, O has 2^r nonzero entries per row. The direct efficiently-computable-sparse-operator interface therefore does not yield a polynomial bound at large r. Changing basis transfers the obligation to amplitude and sampling access for the transformed state.

There is an explicit baseline reduction to the normalized Pauli-average machinery of Zhang and Zhang [6, Section IV.1, Algorithm 1]. Expand $D_Q = \sum_l \alpha_l Z_l$ with $\alpha_l = 2^{-n} W_Q(l)$, conjugate by F_J, and multiply by Z_e. The support invariance proved above ensures the relevant Paulis commute with Z_e and are Hermitian. There are 2^r coefficients of magnitude $2^{-r/2}$; their absolute sum is $2^{r/2}$. Estimating the normalized signed average to error $\epsilon/2^{r/2}$ gives a generic $2^r\epsilon^{-2}\log(1/\delta)$ sample bound. This is an upper bound for that reduction, not a lower bound for their framework.

Finally, Bravyi et al. [2, Section 4.3] supply the norm primitive and linear decomposition-length cost. Our integration constructs the needed states and Clifford access and combines the sector and norm randomness without multiplying two inverse-error costs. We claim a constructive source-specific theorem, not priority for its ingredients, an exhaustive non-subsumption result, or best-known performance.

8 Reproducible finite evidence

The companion code uses Python's standard library and exact integers and rational numbers. The check suite independently enumerates 216 $\mathbb{Z}_4$ sums and 12,240 affine evaluations; constructs ten full small sector/operator/norm identities; checks 274 production norm values against dense vectors; and exhausts 32,962 equatorial samples across ten exact conditional moment identities. It includes a zero-norm sector, nonzero offsets and inputs, hidden coordinates, and two- and three-control cases. Eight full native-source runs compare both readouts with independent exact Fourier-state

evolution; small cases additionally replay literal gates. During the new readout the old 4^k branch routine is forbidden by the test.

Source	n	r	k	p	New calls	Pair calls
Determinant, $t = 2$	10	6	1	5/32	7,168	128
Hidden complement	10	6	1	27/32	7,168	128
Mixed, nonzero input	10	6	1	25/64	7,168	128
Rank four, nonzero input	5	4	1	1/2	28,672	512
Matching offset	4	2	0	3/4	3,584	32
Exact shortcut	3	0	0	1/4	1	1
Two controls	10	8	2	21/64	14,336	512
Three controls	11	10	3	1/2	28,672	2,048

Table 2: Executed matched tasks at $h = 2$, $b = 1$ except the rank-four row at $b = 2$. Counts are real quadratic-sum calls, not equal-cost machine operations. Exact references are shown; sampled answers and seeds are in the receipt.

For $h = 2$ the new-to-pair Gaussian call ratio is $112/2^k$. Thus the executed $k = 1, 2, 3$ cases favor the retained pair count by factors $56, 28, 14$. New widths $r - k, 2r - k$ also differ from the pair width $2r - 2k$. The arithmetic crossover of these call budgets at $k \geq 7$ is not a measured speed crossover. The receipt separately records shared source/discovery, extra preparation and readout timings as single local observations, not a performance benchmark. New full norm runs reach 11 qubits. Analytic family parameters $(n, r) = (26, 18), (50, 38), (82, 66)$ are not claims of new norm runs at those sizes.

Twenty-eight tests at $b = 1, 7, 257, 4096$ inspect exact schedules and verify refusal before drawing with a zero sample budget. They do not execute 4^b samples at high precision. Two further cases retain explicit discovery/grammar refusals. Finite seeded successes test the implementation; Theorems 3.2 and 4.2, not those successes, supply the probability bounds.

9 Publication status and formalization boundary

This candidate is an informal methods preprint. It has undergone internal mathematical and executable checks with AI assistance, not independent peer review. Lean certification has not started. The first manuscript publication and verification and application of its paper DOI precede any formalization work. The later target is a statement-by-statement correspondence for the Walsh and sector algebra, discovery, estimator and full cost contract, and family bounds. Certification must distinguish proved declarations from imported hypotheses and must not be inferred from a build or DOI alone. No Python implementation certification is presently claimed.

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The method excludes unrecognized gate words and sources without its admitted controls or discovery promise. Its certificate bounds are not classical simulation lower bounds. There is no hardware experiment, quantum-advantage claim or consequence resolving BPP versus BQP. More general Fourier-access constructions remain outside this frozen manuscript.

A Evidence and reproduction map

The public package contains the following claim-to-artifact mapping. Stable LaTeX labels in the source, rather than page numbers, identify the intended later formal coverage. All supporting code required for these commands is included; private research links are provenance only.

Claim	Artifact or check
Lemmas 2.1,3.1; Proposition 2.2	`src/sector_cubic_sampling.py` and source reduction dependencies
Theorem 3.2	Verified contraction/exterior search in that module
Lemma 4.1; Theorem 4.2; exact norm arithmetic	`src/sector_norm_sampling.py`; independent exhaustive checker
Section 5, Table 2	`artifacts/s175-sector-norm-checks.json` with descriptors and budgets
Proposition 6.1; prior-work relations	Self-contained proofs here; exported comparison notes and bibliography

From the companion repository root, run

```
python src/check_sector_norm_sampling.py
```

This regenerates the finite receipt, including machine-dependent timings. Compare mathematical fields and canonical-LF source hashes, not timing bytes. To build this PDF, run Tectonic on `manuscript/main.tex`, or use a LaTeX/BibTeX toolchain. The package's `REPRODUCING.md` records the exact tested commands and environment; `PROVENANCE.json` records origin commits and hashes; `MANIFEST.json` records distributed artifact hashes. `SOURCES.md`, `REVIEW.md` and `INTEGRITY.md` disclose sources, checks, AI assistance and claim boundaries. The technical baseline is commit `75694fa51add1c5293168349c4d29af9ef4030ea`; the integrated norm and comparison evidence is `9dc9391292e31c8b6fafb5b53db47899974a858a` in the research repository.

References

[1] Markus Bläser, Michael Gullans, Dominik Hangleiter, Yuxuan Liu, and Youming Qiao. Verifiable quantum advantage based on polynomials with planted structures. arXiv:2609.39918v1, 2026.

[2] Sergey Bravyi, Dan Browne, Padraic Calpin, Earl Campbell, David Gosset, and Mark Howard. Simulation of quantum circuits by low-rank stabilizer decompositions. *Quantum*, 3:181, 2019. Version consulted: arXiv:1808.00128v2.

[3] Dmitri Maslov, Sergey Bravyi, Felix Tripier, Andrii Maksymov, and Joe Latone. Fast classical simulation of Harvard/QuEra IQP circuits. arXiv:2402.03211v1, 2024.

[4] Guedong Park, Hyukjoon Kwon, and Hyunseok Jeong. Extending classically simulatable bounds of Clifford circuits with nonstabilizer states via framed Wigner functions. *Physical Review Letters*, 133:220601, 2024. Version consulted: arXiv:2307.16688v2.

[5] Maarten Van den Nest. Simulating quantum computers with probabilistic methods. *Quantum Information and Computation*, 11(9–10):784–812, 2011. Version consulted: arXiv:0911.1624v3.

[6] Yifan Zhang and Yuxuan Zhang. Classical simulability of quantum circuits with shallow magic depth. arXiv:2409.13809v2, 2025.